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SUMMARY

Robotics and controls engineer, 4+ years taking safety-critical medical automation from R&D to production. Real-time control at 1 kHz, signal processing, computer vision and embedded C++ on regulated hardware. MS Robotics and Controls, Columbia (ROAM Lab).

EXPERIENCE

Clarapath Inc. — Robotics and Controls Engineer — Hawthorne, NY — Apr 2022 – Present

Controls lead, 6-person controls and AI group, mentoring and onboarding new hires and interns, collaborated with pathologists/histologists. Product: a robotic dual microtome system that slices tissue microns thin onto slides using gantries and a tape system (web handler).

- **Built** trajectory optimization for microtome paths using control barrier functions, nonlinear optimization and Bezier curves, under safety constraints - process time 2x faster (1.6s -> 0.8s) [MATLAB, Python, C++]
- **Introduced** cut-signal processing for time/block invariance using PCHIP resampling and Welford statistics, real-time error detection and feature extraction; peak detection on undersampled signals to measure section and block characteristics for user and consumable safe operation [C++, Python]
- **Led** a cross-functional root-cause investigation on section quality; conducted multiple design-of-experiments to isolate root cause, then the mechanical redesign and control rework to run without it — defect rate 19% to under 1% [MATLAB, Python]
- **Built** the product's data-analysis capability from scratch: contact detection and closed-loop depth estimation at the blade to refine block-level cutting; segmentation of block/tissue regions [Python, OpenCV, PyTorch]
- **Owned** controls and embedded software for 10+ subsystems on a multi-node ROS architecture — real-time signal processing, 1 kHz adaptive control, finite state machines, and 0.25 mm laser landmark calibration (plane-normal fit, Rodrigues rotation to CAD nominal) and motion planning for 3-DOF gantries, microtome and tape driver - reduced production time and cost by 3x [C++, SQL, Python]
- **Designed** the machine-to-server telemetry stack for continuous data logging/plotting from fielded units with safe file-handling [C++, SQL]
- **Built** power-on self-test sequences, firmware versioning and acceptance test protocols to statistically validate builds on the manufacturing floor [Python, SQL]
- **Introduced** user-independent inter-subsystem processes, e.g. block-to-tape pressing, with closed-loop control, image analysis and calibration processes removing build variation tuning and improving deliverable quality [C++, Python]
- **Designed** the error-handling architecture of multi-layer error codes and loggers to a <1% field error rate; introduced dependency-ordered safe-mode entry and initialization; raised machine throughput 1.6x [C++, Bash, Python]
- **Enforced** analytical sizing and tolerance analysis before fabrication instead of build-and-test — 2x fewer design iteration cycles
- **Devised** innovative solutions by means of linearized non-linear controls, Fourier analysis, grid search and signal filters enabling machine critical functionality while reducing mechanical redesign costs [MATLAB, Python, C++]

Columbia University, ROAM Lab (Dr. Matei Ciocarlie) — Graduate Research Assistant — New York, NY — Nov 2020 – Jan 2022

- **Built** controls firmware for a modular biomimetic serpent robot, applying model-free reinforcement learning policies, model predictive control and PID loops to locomotion and state estimation [Python, C++, ROS]
- **Integrated** IMUs, magnetic encoders and optical sensors over a CAN network at 100 Hz, logging into ROS nodes [C++]

Evtek — Machine Learning Research Intern — New York, NY — Sep 2021 – Dec 2021

- **Deployed** deep learning models for identification, masking and sorting in automated recycling, and improved conveyor object recognition and tracking 40% using Kalman filtering [OpenCV, PyTorch]

EDUCATION

Columbia University, School of Engineering and Applied Science — New York, NY — Dec 2021

MS Mechanical Engineering (Robotics and Controls), GPA 3.8. TA: Controlled Dynamic Systems, ML for Mechanical Systems.

New York University, Tandon School of Engineering — Brooklyn, NY — May 2020

BS Mechanical Engineering, cum laude. GPA 3.6. Minors in Aerospace Engineering, Nuclear Engineering and Computer Science.

SELECTED PROJECTS

- **Robotics algorithms** (github.com/geojob): RRT and MoveIt motion planning, Cartesian control and inverse kinematics on UR5 and KUKA arms, particle-filter state estimation, model-free RL and evolutionary algorithms [C++, Python, Julia, ROS]

SKILLS

Languages: C++, Python, MATLAB, SQL, Bash, Java, Julia, CUDA, R

Robotics and Controls: ROS, real-time control, nonlinear control, optimal control, adaptive control, model predictive control, control barrier functions, state estimation, sensor fusion, Kalman filtering, motion planning, inverse kinematics, manipulation, Gazebo, MuJoCo, linear algebra, optimization, probability, stochastic processes, dynamics, differential equations, Fourier analysis, design of experiments

Machine Learning and Vision: PyTorch, TensorFlow, JAX, OpenCV, deep learning, reinforcement learning, imitation learning, image segmentation, object tracking

Software and Design: Git, version control, continuous integration, unit test harnesses, Linux, Docker, embedded firmware, SOLIDWORKS, CATIA, ANSYS, Altair, PCB design